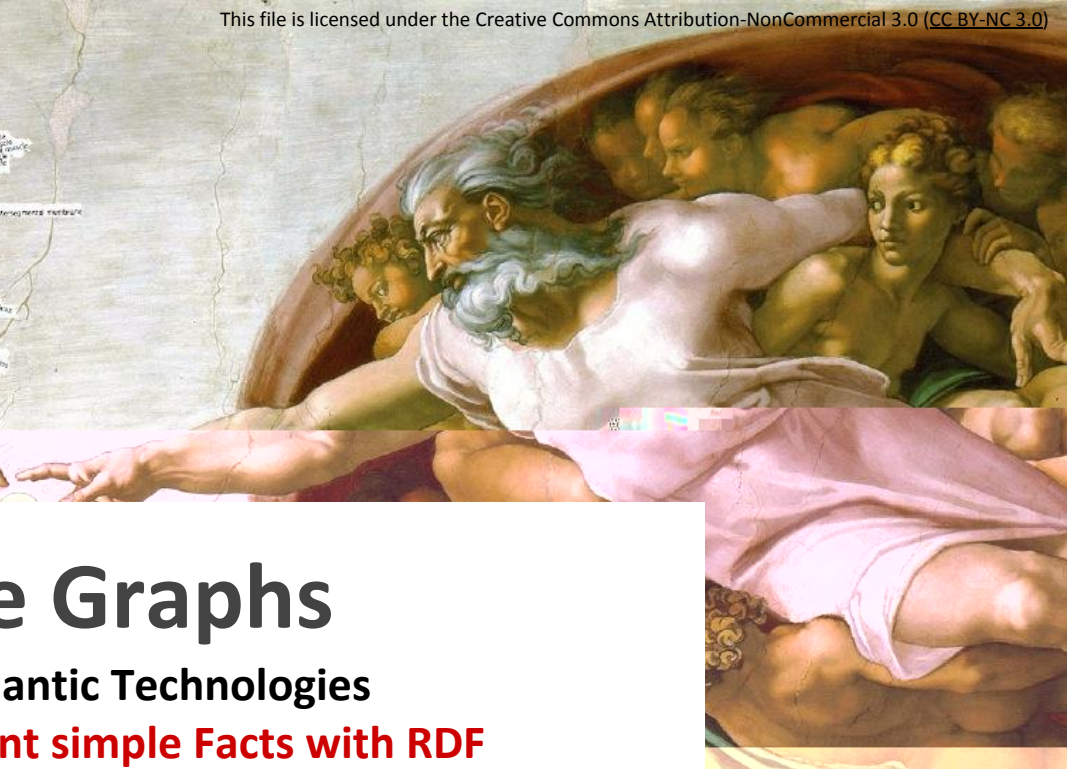
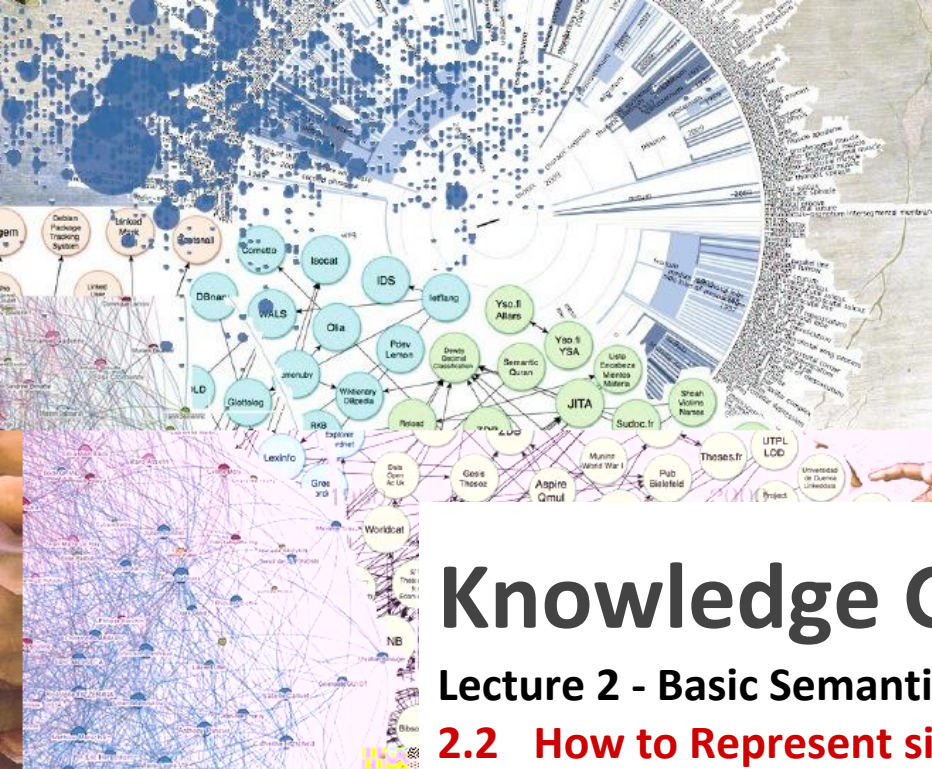




Knowledge Graphs

Lecture 2 - Basic Semantic Technologies

2.2 How to Represent simple Facts with RDF

Prof. Dr. Harald Sack & Dr. Mehwish Alam

FIZ Karlsruhe - Leibniz Institute for Information Infrastructure

AIFB - Karlsruhe Institute of Technology

Autumn 2020



FIZ Karlsruhe
Leibniz-Institut für Informationsinfrastruktur

Knowledge Graphs

Lecture 2: Basic Semantic Technologies

2.1 How to Identify and Access Things

2.2 How to Represent Simple Facts with RDF

2.3 RDF Turtle Serialization

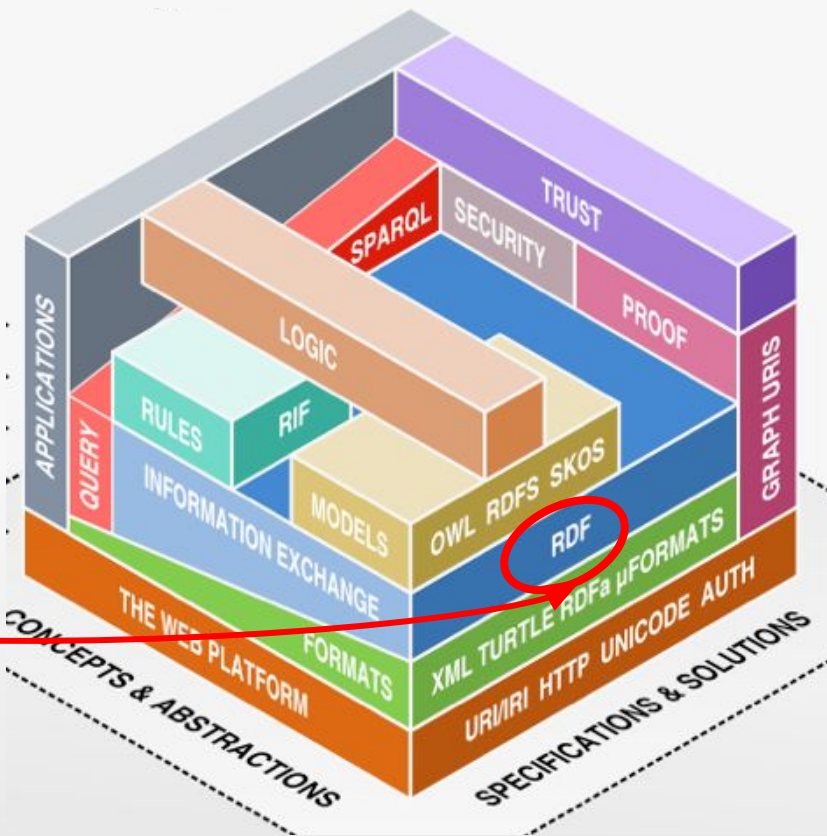
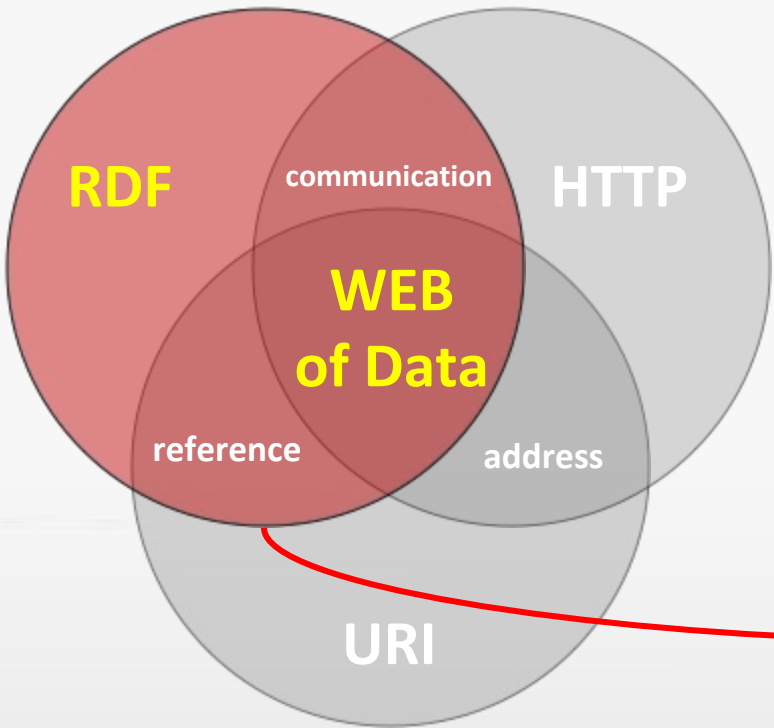
2.4 RDF Complex Data Structures

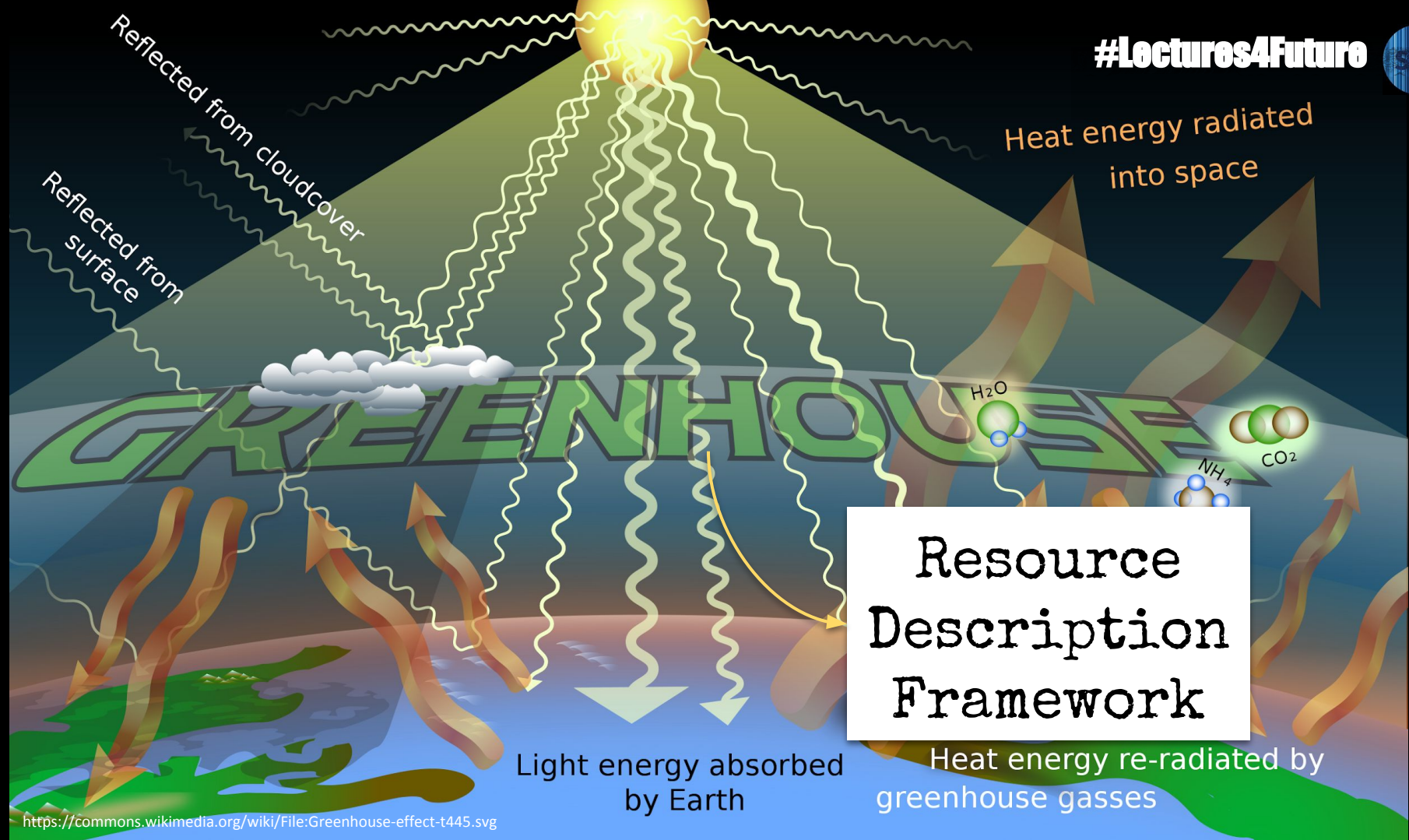
2.5 Model Building with RDFS

2.6 Logical Inference with RDF(S)

Excursion 1: RDFa - RDF and the Web

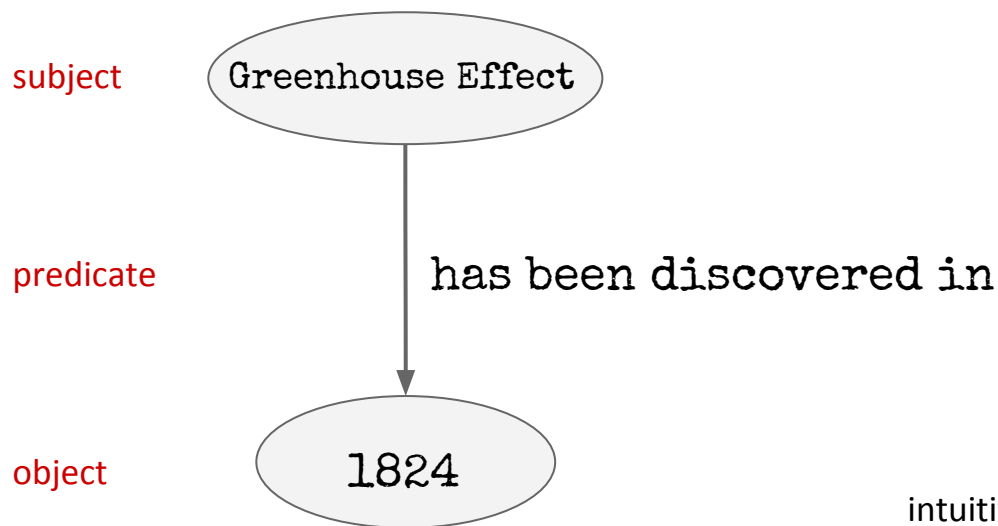
Basic Architecture of the Web of Data





How to represent Knowledge?

- How do I represent the following fact:
“The Greenhouse Effect has been discovered in 1824” in an intuitive way?

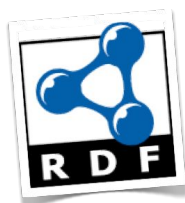


intuitive knowledge representation via a **directed graph**



Resource Description Framework





Resource Description Framework

- **RDF Statements (RDF-Triple):**

Subject
URI

Property
URI

Object / Value
URI / Literal

In RDF the predicate of a statement is referred to as "Property"

N-Triples Serialization

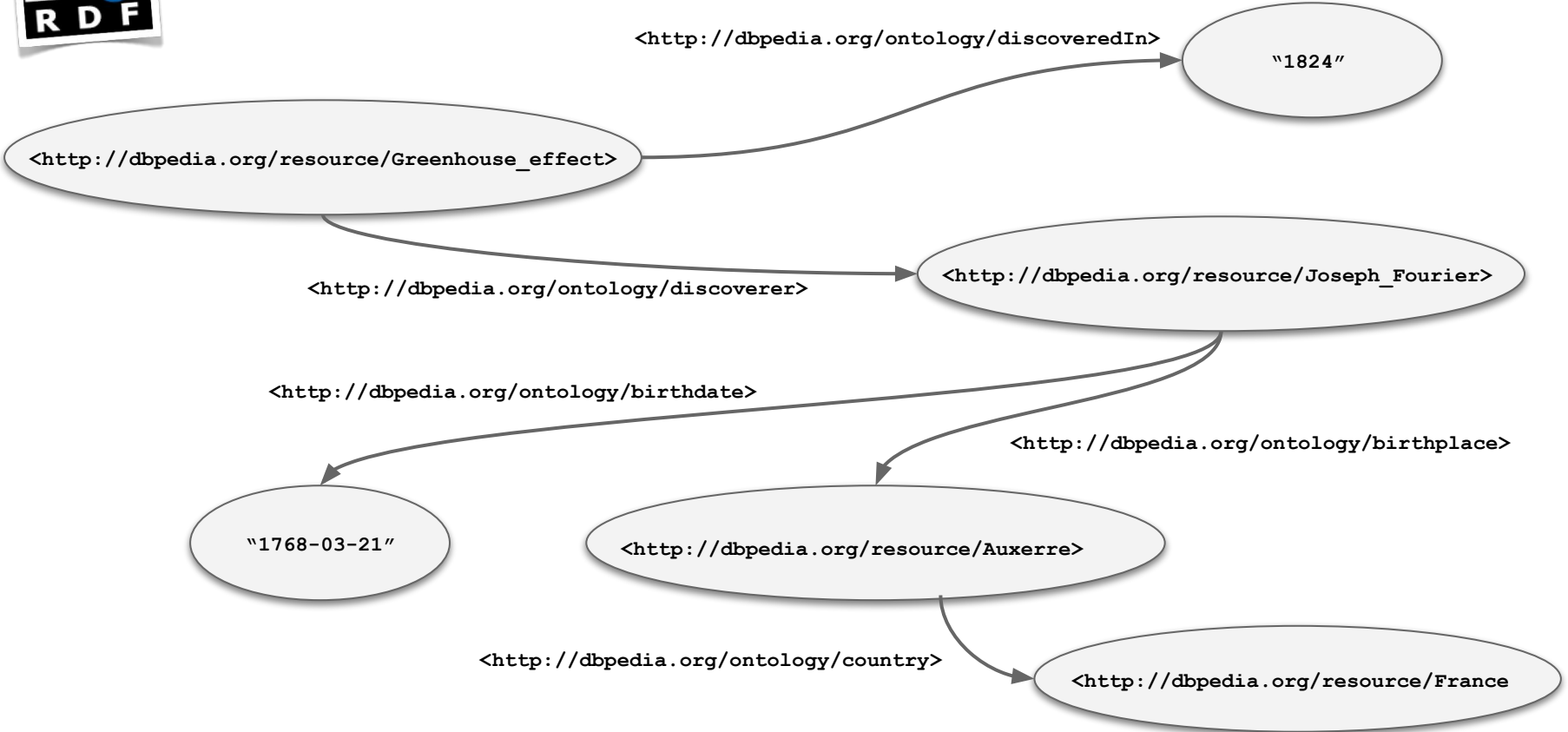
`<http://dbpedia.org/resource/Greenhouse_effect> <http://dbpedia.org/ontology/discoveredIn> "1824" .`



Graph Representation



Resource Description Framework





Resource Description Framework

<code><http://dbpedia.org/resource/Greenhouse_effect></code>	<code><http://dbpedia.org/ontology/discoveredIn></code>	<code>"1824"</code>
<code><http://dbpedia.org/resource/Greenhouse_effect></code>	<code><http://dbpedia.org/ontology/discoverer></code>	<code><http://dbpedia.org/resource/Joseph_Fourier></code>
<code><http://dbpedia.org/resource/Greenhouse_effect></code>	<code><http://purl.org/dc/terms/subject></code>	<code><http://dbpedia.org/category/Climate_change></code>
<code><http://dbpedia.org/resource/Greenhouse_effect></code>	<code><http://purl.org/dc/terms/subject></code>	<code><http://dbpedia.org/category/Athmosphere></code>
...	...	...
<code><http://dbpedia.org/resource/Joseph_Fourier></code>	<code><http://dbpedia.org/ontology/birthdate></code>	<code>"1768-03-21"</code>
<code><http://dbpedia.org/resource/Joseph_Fourier></code>	<code><http://dbpedia.org/ontology/birthplace></code>	<code><http://dbpedia.org/resource/Auxerre></code>
<code><http://dbpedia.org/resource/Joseph_Fourier></code>	<code><http://dbpedia.org/ontology/field></code>	<code><http://dbpedia.org/resource/Physicist></code>
...	...	...
<code><http://dbpedia.org/resource/Auxerre></code>	<code><http://dbpedia.org/ontology/country></code>	<code><http://dbpedia.org/resource/France></code>
<code><http://dbpedia.org/resource/Auxerre></code>	<code><http://www.w3.org/2003/01/geo/wgs84_pos#lat></code>	<code>"47.798599"^^xsd:float</code>
<code><http://dbpedia.org/resource/Auxerre></code>	<code><http://www.w3.org/2003/01/geo/wgs84_pos#long></code>	<code>"3.567200"^^xsd:float</code>
...	...	...
Subject	Property	Object

RDF Triples



Resource Description Framework

<http://dbpedia.org/resource/Greenhouse_effect> <<http://dbpedia.org/ontology/discoveredIn>> "1824" .
<http://dbpedia.org/resource/Greenhouse_effect> <<http://dbpedia.org/ontology/discoverer>> <http://dbpedia.org/resource/Joseph_Fourier> .
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... ..

<http://dbpedia.org/resource/Joseph_Fourier> <<http://dbpedia.org/ontology/birthdate>> "1768-03-21" .
<http://dbpedia.org/resource/Joseph_Fourier> <<http://dbpedia.org/ontology/birthplace>> <<http://dbpedia.org/resource/Auxerre>> .
<http://dbpedia.org/resource/Joseph_Fourier> <<http://dbpedia.org/ontology/field>> <<http://dbpedia.org/resource/Physicist>> .
... ..

<<http://dbpedia.org/resource/Auxerre>> <<http://dbpedia.org/ontology/country>> <<http://dbpedia.org/resource/France>> .
<<http://dbpedia.org/resource/Auxerre>> <http://www.w3.org/2003/01/geo/wgs84_pos#lat> "47.798599"^^xsd:float .
<<http://dbpedia.org/resource/Auxerre>> <http://www.w3.org/2003/01/geo/wgs84_pos#long> "3.567200"^^xsd:float .
... ..

Individuals (Entities)



Resource Description Framework

<http://dbpedia.org/resource/Greenhouse_effect> <http://dbpedia.org/ontology/discoveredIn> "1824" .
<http://dbpedia.org/resource/Greenhouse_effect> <http://dbpedia.org/ontology/discoverer> <http://dbpedia.org/resource/Joseph_Fourier> .
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... ..

<http://dbpedia.org/resource/Joseph_Fourier> <http://dbpedia.org/ontology/birthdate> "1768-03-21" .
<http://dbpedia.org/resource/Joseph_Fourier> <http://dbpedia.org/ontology/birthplace> <http://dbpedia.org/resource/Auxerre> .
<http://dbpedia.org/resource/Joseph_Fourier> <http://dbpedia.org/ontology/field> <http://dbpedia.org/resource/Physicist> .
... ..

<http://dbpedia.org/resource/Auxerre> <http://dbpedia.org/ontology/country> <http://dbpedia.org/resource/France> .
<http://dbpedia.org/resource/Auxerre> <http://www.w3.org/2003/01/geo/wgs84_pos#lat> "47.798599"^^xsd:float .
<http://dbpedia.org/resource/Auxerre> <http://www.w3.org/2003/01/geo/wgs84_pos#long> "3.567200"^^xsd:float .
... ..

Classes



Resource Description Framework

```
<http://dbpedia.org/resource/Greenhouse_effect> <http://dbpedia.org/ontology/discoveredIn> "1824" .
<http://dbpedia.org/resource/Greenhouse_effect> <http://dbpedia.org/ontology/discoverer> <http://dbpedia.org/resource/Joseph_Fourier> .
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<http://dbpedia.org/resource/Greenhouse_effect> <http://purl.org/dc/terms/subject> <http://dbpedia.org/category/Athmosphere> .
...
...
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<http://dbpedia.org/resource/Joseph_Fourier> <http://dbpedia.org/ontology/birthdate> "1768-03-21" .
<http://dbpedia.org/resource/Joseph_Fourier> <http://dbpedia.org/ontology/birthplace> <http://dbpedia.org/resource/Auxerre> .
<http://dbpedia.org/resource/Joseph_Fourier> <http://dbpedia.org/ontology/field> <http://dbpedia.org/resource/Physicist> .
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<http://dbpedia.org/resource/Auxerre> <http://dbpedia.org/ontology/country> <http://dbpedia.org/resource/France> .
<http://dbpedia.org/resource/Auxerre> <http://www.w3.org/2003/01/geo/wgs84_pos#lat> "47.798599"^^xsd:float .
<http://dbpedia.org/resource/Auxerre> <http://www.w3.org/2003/01/geo/wgs84_pos#long> "3.567200"^^xsd:float .
...
...
...
```

Literals



Resource Description Framework

<http://dbpedia.org/resource/Greenhouse_effect>	<http://dbpedia.org/ontology/discoveredIn>	"1824"
<http://dbpedia.org/resource/Greenhouse_effect>	<http://dbpedia.org/ontology/discoverer>	<http://dbpedia.org/resource/Joseph_Fourier>
<http://dbpedia.org/resource/Greenhouse_effect>	<http://purl.org/dc/terms/subject>	<http://dbpedia.org/category/Climate_change>
<http://dbpedia.org/resource/Greenhouse_effect>	<http://purl.org/dc/terms/subject>	<http://dbpedia.org/category/Athmosphere>
...	...	...
<http://dbpedia.org/resource/Joseph_Fourier>	<http://dbpedia.org/ontology/birthdate>	"1768-03-21"
<http://dbpedia.org/resource/Joseph_Fourier>	<http://dbpedia.org/ontology/birthplace>	<http://dbpedia.org/resource/Auxerre>
<http://dbpedia.org/resource/Joseph_Fourier>	<http://dbpedia.org/ontology/field>	<http://dbpedia.org/resource/Physicist>
...	...	...
<http://dbpedia.org/resource/Auxerre>	<http://dbpedia.org/ontology/country>	<http://dbpedia.org/resource/France>
<http://dbpedia.org/resource/Auxerre>	<http://www.w3.org/2003/01/geo/wgs84_pos#lat>	"47.798599"^^xsd:float
<http://dbpedia.org/resource/Auxerre>	<http://www.w3.org/2003/01/geo/wgs84_pos#long>	"3.567200"^^xsd:float
...	...	...

Properties



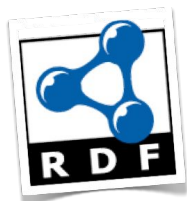
Resource Description Framework

<http://dbpedia.org/resource/Greenhouse_effect> <<http://dbpedia.org/ontology/discoveredIn>> "1824" .
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<http://dbpedia.org/resource/Joseph_Fourier> <<http://dbpedia.org/ontology/birthplace>> <<http://dbpedia.org/resource/Auxerre>> .
<http://dbpedia.org/resource/Joseph_Fourier> <<http://dbpedia.org/ontology/field>> <<http://dbpedia.org/resource/Physicist>> .
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... ..

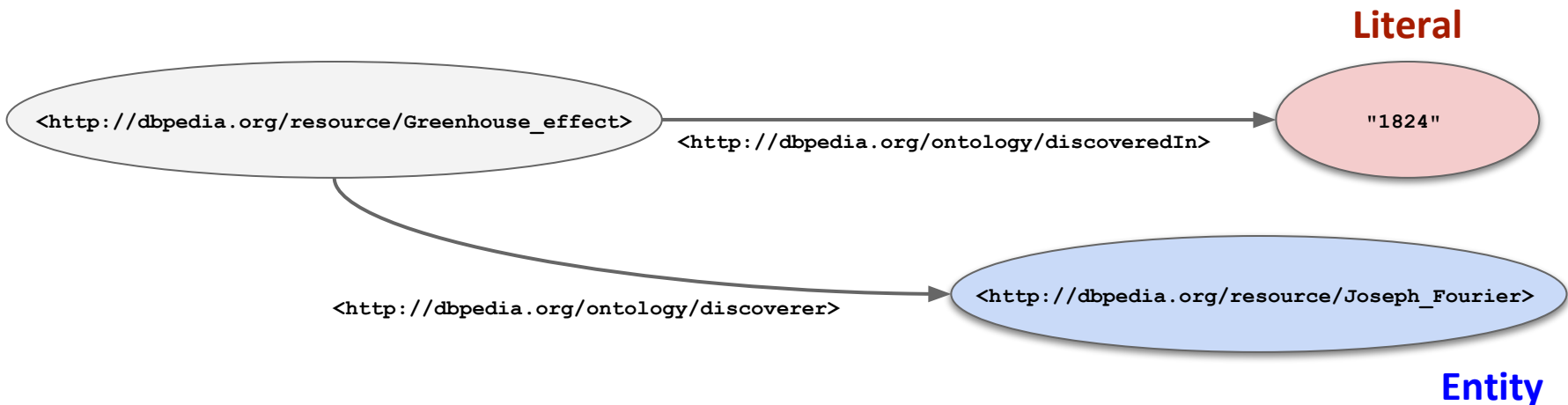
Vocabularies / Ontologies



Resource Description Framework

- **URIs and Literals**

- **URIs** identify and reference resources uniquely.
- **Literals** describe data values that don't have a separate existence.



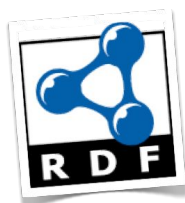


RDF Literals and Datatypes

- Typed literals can be expressed via **XML Schema datatypes**.
- Namespace for typed literals:
`http://www.w3.org/2001/XMLSchema#`
- Examples:
`"Semantics"^^<http://www.w3.org/2001/XMLSchema#string>`
`"1161.00"^^<http://www.w3.org/2001/XMLSchema#float>`
`"2015-08-02"^^<http://www.w3.org/2001/XMLSchema#date>`
- **Language Tags** denote the (natural) language of the text:
 - Example:
`"Semantik"@de , "Semantics"@en`

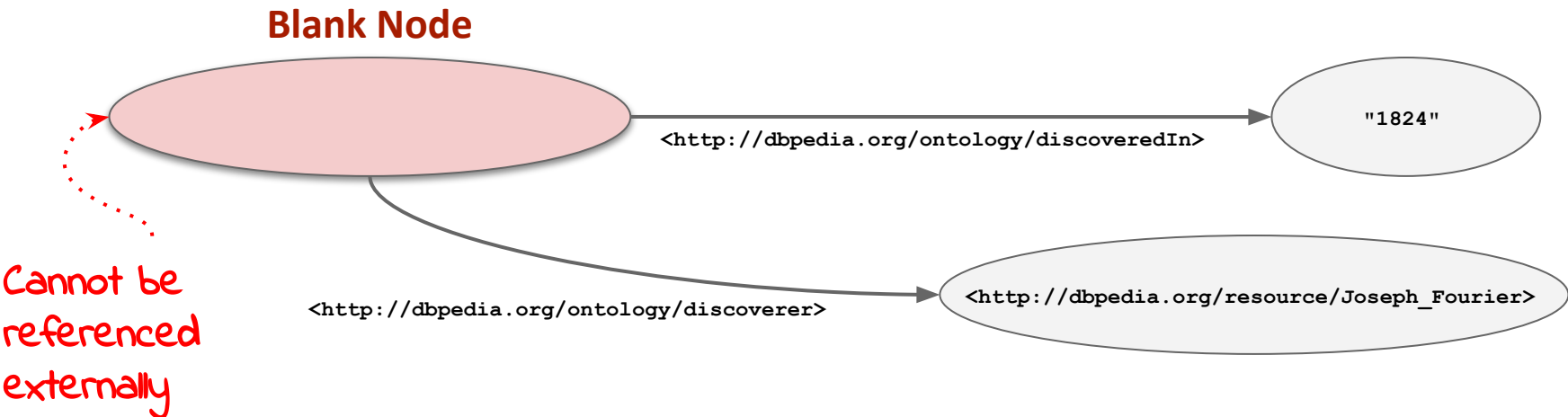
<http://www.w3.org/TR/2013/WD-rdf11-concepts-20130115/#xsd-datatypes>

	xsd:string	Character strings
	xsd:boolean	true, false
	xsd:decimal	Arbitrary-precision decimal
	xsd:integer	Arbitrary-size integer numbers
IEEE floating-point numbers	xsd:double	64-bit floating point numbers
	xsd:float	32-bit floating point numbers
	xsd:date	Dates (yyyy-mm-dd) with or without time
	xsd:time	Times (hh:mm:ss.sss...) with or without time zone
Time and date	xsd:dateTime	Date and time with or without time zone
	xsd:dateTimeStamp	Date and time with required time zone
	xsd:gYear	Gregorian calendar year
	xsd:gMonth	Gregorian calendar month
	xsd:gDay	Gregorian calendar day of the month
	xsd:gYearMonth	Gregorian calendar year and month
	xsd:gMonthDay	Gregorian calendar month and day
	xsd:duration	Duration of time
	xsd:dayTimeDuration	Duration of time (days, hours, minutes, seconds)
	xsd:timeDuration	Duration of time (minutes, seconds)
	xsd:dateTimeDuration	Duration of time (days, hours, minutes, seconds)
	xsd:base64Binary	Base64-encoded binary data
	xsd:hexBinary	Hex-encoded binary data
Limited-range integer numbers	xsd:int	-2147483648...+2147483647 (32 bit)
	xsd:short	-32768...+32767 (16 bit)
	xsd:byte	-128...+127 (8 bit)
	xsd:nonNegativeInteger	0... (unbounded)
	xsd:positiveInteger	1... (unbounded)
	xsd:negativeInteger	-1... (unbounded)
	xsd:nonPositiveInteger	0... (unbounded)
	xsd:integer	-2147483648...+2147483647 (32 bit)
	xsd:long	-9223372036854775808...+9223372036854775807 (64 bit)
	xsd:unsignedInt	0...255 (8 bit)
	xsd:unsignedShort	0...65535 (16 bit)
	xsd:unsignedByte	0...255 (8 bit)
	xsd:float	0...4294967295 (32 bit)
	xsd:double	0...18446744073709551615 (64 bit)
	xsd:decimal	Integer numbers >0
	xsd:integer	Integer numbers ≥0
	xsd:short	Integer numbers <0
	xsd:byte	Integer numbers ≤0
Encoded binary data	xsd:base64Binary	Base64-encoded binary data
	xsd:hexBinary	Hex-encoded binary data
	xsd:uri	Absolute or relative URIs
	xsd:language	Language tags per [BCP47]
	xsd:normalizedString	Whitespace-normalized strings
	xsd:string	Tokenized strings
	xsd:token	XML NMTOKENs
	xsd:NCName	XML Names
	xsd:Name	XML NCNames



RDF Blank Nodes

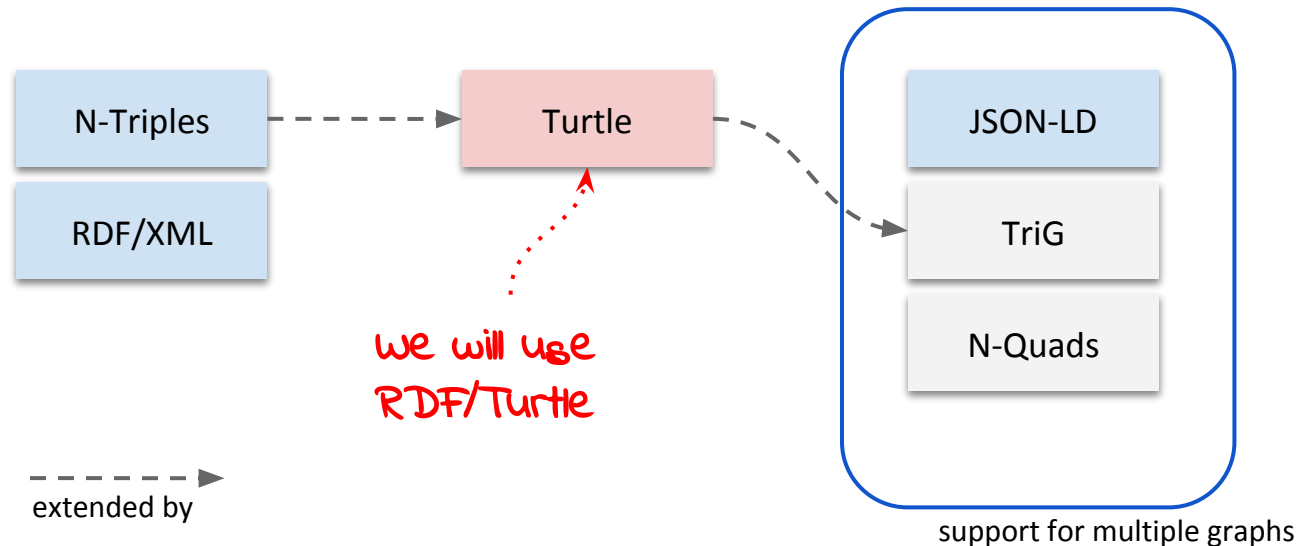
- Blank Nodes
 - denote **existence of an individual** with specific attributes, but **without providing an identification or reference**.

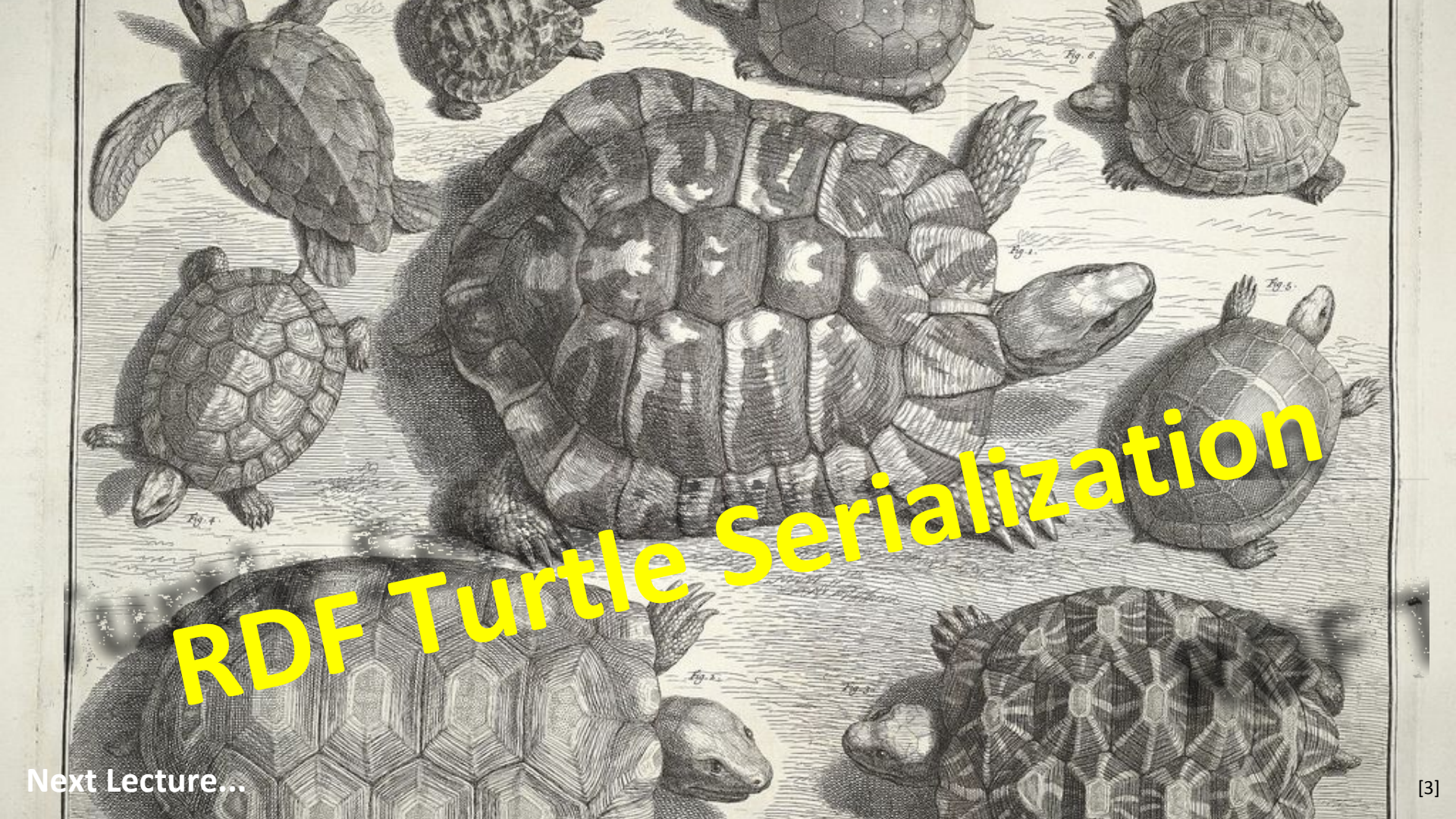




RDF Serializations

- RDF comes with several different **serialization formats**:
 - N-Triples, RDF/XML, JSON, Turtle, TriG, N-Quads, RDFa, ...





RDF Turtle Serialization

Next Lecture...

Picture References:

- [1] Benjamin Nowack, *The Semantic Web - Not a Piece of cake...*, at bnode.org, 2009-07-08 , [CC BY 3.0]
<http://bnode.org/blog/2009/07/08/the-semantic-web-not-a-piece-of-cake>
- [2] The Green House Effect, A loose necktie [CC BY-SA]
<https://commons.wikimedia.org/wiki/File:Greenhouse-effect-t445.svg>
- [3] Albertus Seba Thesaurus Tab. LXXX, 18th century, Albertus Seba [Public domain]
https://commons.wikimedia.org/wiki/File:Albertus_Seba_Thesaurus_Tab._LXXX.jpg